another view of integer multiplication

I want to explore the natural numbers and the polynomial rings via a formalist philosophy that is not set-theoretical, adding operations and relations and seeing what objects we came up with. I will aim towards representing $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ and polynomial rings of these; the formalism of what happens next is well codified in standard undergraduate. The aim is not to get there directly, but to explore on our way there.

The main relations of interest will be:

- existence: imposing the existence of an element, and then if any operation is added imposing (formal) completion.
- identification: imposing $x * x = x$
- virtualization: imposing the existence of an inverse pivoted around an identity. Why is it natural to virtualize when there is an identity; delete this when you figured it out
- element-wise absorption law: given two separate elements x, y is $xy = x$ or $xy = y$

We shall always assume:

- associativity: This is because natural numbers and functions can both have their multiplication and compositing broken down into "blocks" (see ref:HERE for the intuition). As one of the ultimate goals here is to have a better intuition on the primes ideals of $\mathbb{Z}$ and $k[x]$, if we don't impose associativity we are asking the "wrong" question.
- commutativity: lack of associativity means that we can't break down our problem and would have to examine each pair. But with associativity we can think of a sequence of operations being performed in order (where we could "simplify" not needing to "collapse" in sequence but can collapse any two in the chain until we get the answer) and so it is well-defined to write:

$$a_1 * a_2 * \cdots * a_n$$

  Now, lack of commutativity means that order matters when combining two elements. Commutativity represents an "invariance" to direction. If we have associativity and commutativity, then we can think of it as combining operations that is not sensitive to order. Again, if order mattered we would be dealing with something elem so we impose this too.
- distributivity: When there is more than one binary operation, they can either act "independently" from each other giving multiple new elements using formal combinations or (what is naturally more interesting) we want them to meaningfully interact. For an example of what happens if we are not careful, Define $+$ and $\cdot$ on $\{0, 1\}$ so that $0 + 0 = 0$ and $1 + 1 = 1$, but $0 \cdot 0 = 1$ and $1 + 1 = 0$.

  There are many choices that can be made for combining binary operations depending on what we are modelling. In lattice theory, the absorption law is a natural choice. When function addition and composition are taken on affine linear functions, right distributivity is natural.

As we are assuming associativity and commutativity, for any object $(S, +)$ and any $*$ (not necessarily distributive), then we have $(S, +, *) \to (\mathrm{Mor}(S, +), \circ)$ which can be done using regular (left or right) representation. We shall almost always assume this is faithful, meaning there are no annihilators (which is automatic if there is an identity). What distributivity guarantees us is that we can insure the endomorphisms are *linear*[1].

we want faithful reps. If there is an id then automatically, If not we must not have "annihilators". Think about whether you need to consider this.

As $\mathbb{N}$ has this property and we can internet multiplying numbers as getting integral boxes, we shall desires this linearity. We shall think about what it means to have distributivity in the case of polynomial rings, especially since $x \cdot (-)$ is so important for finding char and min polynomials, not to even mention its use in representation theory.

The following commutative diagram is placed to represent the linearity of End in what may be more familiar categorical language to some readers:

$$\begin{tikzcd}
R \times R \times R \arrow[r, "(\mathrm{id},+)"] \arrow[d, "(\cdot,\cdot)"] & R \times R \arrow[d, "\cdot"] \\
R \times R \arrow[r, "+"] & R
\end{tikzcd}
\qquad
\begin{tikzcd}
(r,s,t) \arrow[r, "(\mathrm{id},+)"] \arrow[d, "(\cdot,\cdot)"] & (r, s+t) \arrow[d, "\cdot"] \\
(r \cdot s, r \cdot t) \arrow[r, "+"] & r \cdot s + r \cdot t = r \cdot (s+t)
\end{tikzcd}$$

- at most 2 binary operations: imposing a third binary operation that distributes with both operations is forcing that the original structure essentially respects a ring homomorphism for every element, usually making the 3rd operation trivial or the identity. Usually, unary operators are added instead (think convolution or d). An exception is the Jacobi identity if we drop associativity and make it anti-commutative. The reason for all of this gets complicated quickly and explores behaviour beyond our intended target of $\mathbb{N}$ and $k[x]$, so we are not going to add it.

After the exploration, I may choose to find some interesting models for the algebraic structures that were found.

1 Doing the Work

Let us start with the existence of a single element ϵ and binary operation suggestively labeled $+$. It will eventually act like usual addition but for now only assume we can create formal elements using ϵ and $+$ under the equivalent of associativity and commutativity. We shall do the usual short hand of labelling

$$\underbrace{\epsilon + \cdots + \epsilon}_{n \text{ times}} = n\epsilon$$

The collection of all these elements can be denoted:

$$S = \{\epsilon, 2\epsilon, 3\epsilon, \ldots\}$$

The set S has no identity element. What we can do is impose $\epsilon + \epsilon = \epsilon$. If we do this, then everything shall simply collapse down to a singleton:

$$S = \{\epsilon\}$$

As there is only one binary operation, we cannot virtualize or absorb distinct elements, so our exploration is exhaustive.

[1] not just affine! affine is right but not left distributive

What is not more interesting is to have a 2nd binary operation with distributivity, suggestively labeled $\cdot$. We take the usual convention of:

$$\underbrace{\epsilon \cdot \ \cdots \ \cdot \epsilon}_{n \text{ times}} = \epsilon^n$$

If we add $\epsilon^2 = \epsilon$, then everything becomes ϵ again, so let's not impose this yet. As $\epsilon + \epsilon = \epsilon$ and we assume linearity (i.e. distributivity):

$$\epsilon^2 = (\epsilon)(\epsilon) = (\epsilon + \epsilon)\epsilon = \epsilon^2 + \epsilon^2$$

Note that we are *not* assuming that there is an (additive or multiplicative) inverse, we have yet to introduce "virtual" objects. Thus, $n\epsilon^i = \epsilon^i$. What about $\epsilon + \epsilon^2$?